

3.2

a) ortonormalní bázi podprostoru $L_0\{1, x, x^2\}$

$$\vec{v}_1 = 1 \quad \vec{u}_1 = \frac{1}{\|\vec{1}\|} = \frac{1}{\sqrt{\langle 1, 1 \rangle}} = \frac{1}{\sqrt{\int_1^5 1}} = \frac{1}{\sqrt{4}} = \frac{1}{2}$$

$$x - \vec{w}_2 = x - \langle \frac{1}{2}, x \rangle \cdot \frac{1}{2} = x - \int_1^5 \frac{1}{2} \cdot x \, dx \cdot \frac{1}{2} = x - 6 \cdot \frac{1}{2} = x - 3$$

$$\vec{u}_2 = \frac{x-3}{\|x-3\|} = \frac{x-3}{\sqrt{\int_1^5 (x-3)^2}} = \frac{x-3}{\sqrt{\frac{13}{3}}} = \frac{\sqrt{3} \cdot (x-3)}{4}$$

$$\begin{aligned} x^2 - \vec{w}_3 &= x^2 - \langle \frac{1}{2}, x^2 \rangle \cdot \frac{1}{2} - \langle \frac{\sqrt{3} \cdot (x-3)}{4}, x^2 \rangle \cdot \frac{\sqrt{3}}{4} \cdot (x-3) = x^2 - \int_1^5 \frac{1}{2} \cdot x^2 \, dx \cdot \frac{1}{2} - \int_1^5 \frac{\sqrt{3}}{4} (x-3) x^2 \, dx \cdot \frac{\sqrt{3}}{4} (x-3) \\ &= x^2 - \frac{62}{6} - 6 \cdot (x-3) = x^2 - 6x + \frac{46}{6} = x^2 - 6x + \frac{23}{3} \end{aligned}$$

$$\vec{u}_3 = \frac{x^2 - 6x + \frac{23}{3}}{\|x^2 - 6x + \frac{23}{3}\|} = \frac{x^2 - 6x + \frac{23}{3}}{\sqrt{\int_1^5 (x^2 - 6x + \frac{23}{3})^2}} = \frac{x^2 - 6x + \frac{23}{3}}{\sqrt{\frac{256}{45}}} = (x^2 - 6x + \frac{23}{3}) \cdot \frac{3\sqrt{5}}{16}$$

$$B = \left(\frac{1}{2}, \frac{\sqrt{3}}{4} (x-3), \frac{3\sqrt{5}}{16} (x^2 - 6x + \frac{23}{3}) \right)$$

b) $\vec{w} = \langle \vec{u}_1, \vec{v} \rangle \vec{u}_1 + \langle \vec{u}_2, \vec{v} \rangle \vec{u}_2 + \dots + \langle \vec{u}_k, \vec{v} \rangle \vec{u}_k$ $\vec{w}$ je ortonormalní projekci vektoru v na W

$$\vec{w} = \langle \frac{1}{2}, \sin x \rangle \cdot \frac{1}{2} + \langle \frac{\sqrt{3}}{4} (x-3), \sin x \rangle \cdot \frac{\sqrt{3}}{4} \cdot (x-3) + \langle \frac{3\sqrt{5}}{16} (x^2 - 6x + \frac{23}{3}), \sin x \rangle \cdot \frac{3\sqrt{5}}{16} (x^2 - 6x + \frac{23}{3})$$

$$\vec{w} = \int_1^5 \frac{1}{2} \sin x \, dx \cdot \frac{1}{2} + \int_1^5 \frac{\sqrt{3}}{4} (x-3) \sin x \, dx \cdot \frac{\sqrt{3}}{4} \cdot (x-3) + \int_1^5 \frac{3\sqrt{5}}{16} (x^2 - 6x + \frac{23}{3}) \sin x \, dx \cdot \frac{3\sqrt{5}}{16} (x^2 - 6x + \frac{23}{3})$$

$$\vec{w} = 0,128 \cdot \frac{1}{2} - 1,493 \cdot \frac{\sqrt{3}}{4} \cdot (x-3) - 0,125 \cdot \frac{3\sqrt{5}}{16} (x^2 - 6x + \frac{23}{3})$$

$$= 0,064 - 0,646x + 1,938 - 0,052x^2 + 0,312x - 0,399$$

$$= -0,052x^2 - 0,334x + 1,603 \quad \underline{a = -0,052 \quad b = -0,334 \quad c = 1,603}$$

c) $\|\sin(x) - (ax^2 + bx + c)\|$

$$\begin{aligned} \hookrightarrow \|\sin(x) - (-0,052x^2 - 0,334x + 1,603)\| &= \sqrt{\langle (\sin(x) - (-0,052x^2 - 0,334x + 1,603)), (\sin(x) - (-0,052x^2 - 0,334x + 1,603)) \rangle} \\ &= \sqrt{\int_1^5 (\sin(x) - (-0,052x^2 - 0,334x + 1,603))^2} = \sqrt{0,101} = \underline{0,318} \end{aligned}$$